MIL-F-0010135C(MU)
16 December 1968
USED IN LIEU OF
MIL-F-10135B
2 March 1955

MILITARY SPECIFICATION

FACEBLANK, FIELD PROTECTIVE MASK

This limited coordination Military specification has been prepared by Edgewood Arsenal based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of Military Specification MIL-F-10135B. It is subject to modification. However, pending its promulgation as a coordinated Military Specification, it may be used in procurement.

1. SCOPE

1.1 Scope. This specification covers a protective mask faceblank molded from a 65-percent natural rubber compound.

1.2 Classification. The faceblank shall be of the following sizes, as specified (see 6.2):

Small (E5-2-764 or E5-2-804)
Medium (E5-2-745 or E5-2-805)
Large (E5-2-821)

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-781 - Steel, Strapping, Flat.
NN-P-530 - Plywood, Flat-Panel.
TT-W-572 - Wood Preservative, Water-Repellent.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.

FSC 4240

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STANDARDS

FEDERAL

FED-STD-601 - Rubber: Sampling and Testing.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

E5-2-764 - Faceblank, Small.
E5-2-804 - Faceblank, Small.
E5-2-745 - Faceblank, Medium.
E5-2-805 - Faceblank, Medium.
E5-2-821 - Faceblank, Large.
E136-19-14 - Q38 Natural Rubber Puncture Tester.
D136-19-22 - Q37 Natural Rubber Flexibility Tester.
D150-2-1 - Marking Diagram for Shipping Containers.

PUBLICATIONS

EDGEWOOD ARSENAL DIRECTIVE

No. 618 - Ozone Resistance Test of Natural Rubber Material.

(Copies of specifications, standards, drawings, and publications required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following publications form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal, shall apply.

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UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies of this publication should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago, Illinois 60606.)

AMERICAN TRUCKING ASSOCIATIONS, INC.

National Motor Freight Classification Rules

(Application for copies should be addressed to the American Trucking Associations, Inc., 1616 P Street, N. W., Washington, D. C. 20036.)

SCHWARTZ, TULIPAN, AND BIRMINGHAM

Occupational Diseases of the Skin

(This publication may be obtained from the publishers: Lea and Febiger, Philadelphia, Pennsylvania.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 Rubber compound. The rubber compound shall be nonblooming, except for a slight wax-type bloom, and shall contain no less than 65 percent, by volume, of new natural rubber. The compound shall be free from lead or mineral sulfates.

3.1.2 Antioxidants and accelerators. The rubber compound shall contain only approved antioxidants and accelerators (see 3.1.6).

3.1.3 Antiozonide. The rubber compound shall contain an antiozonide.

3.1.4 Copper and manganese inhibitor. The rubber compound shall contain a copper and manganese inhibitor.

3.1.5 Coloring materials. The rubber compound shall be colored by carbon black as required.

3.1.6 Toxicity. When the faceblank is worn as a component of a protective mask, it shall not cause undue sensitization or primary irritation of the wearer's skin. Objective evidence shall be submitted in adcordance with 4.1.2.2. Once the rubber formulation has been approved by the Government, the supplier shall not make a major change (6.4) in the formulation or in the curing process.

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3.2 Manufacture. The faceblank shall be manufactured in one piece by the conventional process of molding under heat and pressure. Each size specified in the contract or order shall be manufactured in accordance with the applicable drawing listed in 2.1 in molds approved by the Government.

3.2.1 Repair of flaws. Faceblanks with small flaws (major or minor defects) may be repaired and accepted provided the repair is vulcanized with material of the same approved formulation as the faceblank and the repaired area meets the requirements of physical properties of 3.3. The appearance of the repair shall be such that the color and shape blend into the surrounding area with no break in surface continuity and no feather edges.

3.2.2 Test slabs. Test slabs 6 by 6 inches, by 0.075 ± 0.01 inch shall be made from stock representative of the formulation used for the preproduction sample and from each sublot during regular production of faceblanks. The slabs shall have a cure equivalent to that of the regular production faceblank (see 6.3).

3.3 Physical properties. Physical properties shall conform to the requirements shown in table 1 when tested as specified therein.

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Table 1. Physical properties

Property	Unaged	After heat aging	Test paragraph
Average tensile strength, psi, min	3000	2550	4.4.4.2.1
Average ultimate elongation, %, min	600		4.4.4.2.1
Percent increase, max		20	
Percent decrease, max		25	
Average tensile stress at 200% elongation, psi	250 to 450	250 to 450	4.4.4.2.1
Individuals, psi, min	240	240	
Average tear resistance, lbs/in., min			4.4.4.2.1
Die A or B	300	210	
Die C	225	165	
Average tension set at 75% ultimate elongation, %, max	20		4.4.4.2.1
Average hardness (Shore "A" or "A2")	47 ± 3		4.4.4.2.1
Ozone resistance, min	80		4.4.4.2.2
Flexibility			4.4.4.2.3
Units at 75° ± 5°F, min	7.5		
Units retained at -25° ± 5°F, min	4.8		
Puncture resistance, psi, min	9.5		4.4.4.2.4

3.4 Preproduction. Prior to the start of regular production, a preproduction lot of faceblanks shall be produced for examination and testing in accordance with 4.3. Any change in formulation, manufacturing methods, or molds shall require a new preproduction lot.

3.5 Critical areas. The peripheral sealing area contacting the wearer's skin, the canister mounting area, and the lens sealing surface shall be considered as critical areas of the faceblank.

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3.6 Workmanship. The faceblank inner and outer surfaces shall be smooth and free from imperfections such as pits, bumps, porous areas, cracks, flash, abrasions, checks, blisters, or foreign matter. The rubber shall be compact and free from holes and bubbles. Ridges of mold-parting lines on outer surfaces shall not be considered as flash when closely trimmed. Rips, tears, or punctures shall be considered as damage. Workmanship defects of a magnitude greater than the referee standards established by the Government for major and minor defects (see 6.7) shall be categorized in the appropriate defect class.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence.

4.1.2.1 Materials. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.1.2.2 Toxicity. Within 30 days after the date of award of a contract, and without additional cost to the Government, the contractor shall submit the information listed below which will constitute his evidence for establishing the nontoxicity of the rubber formulation. The Surgeon General will act as advisor to the contracting officer in evaluating this data for determining if the formulation is acceptable.

(a) Information relating to candidate formulation.

(1) The name of the contractor and the producing subcontractor, if any.

(2) A description of the end item and its components. Copies of applicable specifications will be included.

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(3) The number of items under procurement for which the proposed formulation will be used.

(4) A description of the candidate formulation that will include:

a. The chemical and trade names of the ingredients.

b. The quantity of each ingredient, including the range of variation that can be tolerated without loss of quality in the finished product. Quantities shall be expressed in percent by weight and parts by weight.

c. The percent of rubber by volume.

d. The mold lubricants, dusting powders, or wet sprays proposed.

e. Proposed cure times and temperature.

f. Results of previous toxic evaluation demonstrating acceptability or evidence of prior government approval of the proposed formulation.

g. Information and data to substantiate the nontoxicity of the product. This information will include a complete description of any programs the contractor has carried out to prove nontoxicity, such as prior patch testing with unlined items the use of which involves direct contact with the skin; evidence that the proposed formulation is the same as that used in his commercial products; and information on the nature, extent, and length of time of his commercial experience with the formulation. Other evidence may include references to technical or other scientific publications, commercial laboratory reports, opinions or statements of recognized authorities and agencies in the field of dermatology, and any other pertinent data.

(b) Patch testing. If the above information is not sufficient to establish nontoxicity of the rubber formulation, the contractor shall perform 200 patch tests under the supervision of competent medical authorities in the field of dermatology. These patch tests shall be at no additional cost to the Government and shall be conducted in accordance with the prophetic patch test method described in "Occupational Diseases of the Skin" by Schwartz, Tulipan, and Birmingham. A control rubber for use in the patch tests will be furnished by the Government upon written request of the contractor. The patches composed of the candidate rubber formulation shall not produce positive reactions in excess of that of the control rubber patches. A copy of the test results and the evaluation thereof by the medical authorities described above shall be furnished to the contracting officer.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of 30 (20 trimmed and 10 untrimmed) faceblanks from each mold cavity and 16 test slabs shall be manufactured using the same methods, materials, equipment, and rubber formulations as will be used during regular production.

4.3.2 Inspection procedure.

4.3.2.1 For examination. Fifteen of the sample faceblanks shall be inspected for all requirements of the drawings and this specification.

4.3.2.2 For tests. Fifteen faceblanks and 16 test slabs, produced in accordance with 4.3.1, shall be tested in accordance with 4.4.4.

4.3.3 Tests. Tests shall be conducted as specified in 4.4.4.

4.3.4 Acceptance/rejection criteria. The preproduction sample shall meet the examination and tests specified by 4.3.2.1 and 4.3.2.2 to be acceptable. The supplier shall obtain written approval from the contracting officer before proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the faceblanks of one size produced in sequence during one month as indicated by the dots in the medallion on the faceblank. A sublot shall consist of the faceblanks produced from one batch of material (see 6.5). A new lot shall be established following the occurrence of any event affecting the homogeneity of the lot or use of an alternative approved formulation.

4.4.2 Sampling.

4.4.2.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For tests. Fifteen faceblanks and 16 test slabs shall be submitted with each sublot for testing.

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4.4.3 Inspection procedure.

4.4.3.1 For examination. The sample faceblanks from the regular production lot shall be examined in accordance with the classification of defects and with MIL-STD-105.

4.4.3.2 For tests. Sample faceblanks and test slabs shall be tested in accordance with 4.4.4. The acceptance number is zero.

4.4.3.3 Classification of defects.

(a) Faceblank. See applicable drawing for size.

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Rubber thickness
102	Porosity or nonfills
103	Deformation or permanent set
104	Flow cracks which enlarge on probing
105	Marking incorrect
106	Workmanship (other than Major 109, see 3.6)
107	Mismatched mandrels (when applicable)
108	Split at register
109	Contamination (grease or oil)
	AQL 0.065 percent defective
110	Imperfections or damage in critical areas (see 3.5)
111	Holes or cuts
Minor:	AQL 2.5 percent defective
201	Excess flash or improper trim
202	Marking missing or illegible
203	Access flap holes improperly located
204	Incorrect size
205	Contamination (other than 109)
206	Color incorrect
207	Tab rivet holes, diameter, piercing, or alignment improper

4.4.4 Tests.

4.4.4.1 Test specimens. Two specimens, one from each cheek, shall be cut from the flat cheek areas of each of 15 sample faceblanks. The specimens shall not be buffed or stretched before testing. Ten specimens, five

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from each cheek, shall be tested for tensile strength, ultimate elongation, and tensile stress before aging and 10 specimens tested after aging. Six specimens, three from each cheek, shall be tested for tension set before aging. Two specimens for tear resistance shall be cut from the forehead section of each of 10 of the above 15 sample faceblanks. Ten specimens, five from each cheek, shall be tested before aging and 10 tested after aging. Enough 1 by 1-inch squares shall be cut from cheek and forehead sections to build up five specimens for the hardness test, each specimen containing enough layers to be no less than 3/16 inch thick. The remaining faceblanks shall be used for retest. Twelve specimens, two from each of six test slabs, shall be taken for ozone resistance tests. Eight specimens, two from each of four test slabs, shall be taken for flexibility testing. Six specimens, one from each of six test slabs, shall be taken for puncture resistance tests. The test specimens selected for heat aging shall be aged in an air oven at a temperature of 158° ± 1.8°F for a continuous period of 168 ± 1 hour or at a temperature of 212° ± 1.8°F for a continuous period of 24 ± 1 hour, as specified in FED-STD-601 (see 6.6).

4.4.4.2 Physical properties tests.

4.4.4.2.1 Tensile strength, ultimate elongation, tensile stress, tear resistance, tension set, and hardness. Tensile strength, ultimate elongation, tensile stress, tear resistance, tension set, and hardness shall be determined as specified in FED-STD-601. The angle-type or crescent-type die shall be used for tear resistance testing.

4.4.4.2.2 Ozone resistance. Ozone resistance shall be determined in accordance with Directive No. 618.

4.4.4.2.3 Flexibility. Flexibility shall be determined using the Q37 Natural Rubber Flexibility Tester (E136-19-22). Flexibility shall be read on the grid of the apparatus. Four specimens shall be tested at 75° ± 5°F and four specimens shall be tested at -25° ± 5°F after being conditioned for 72 hours at that temperature.

4.4.4.2.4 Puncture resistance. Resistance to hydrostatic pressure shall be determined using the Q38 Natural Rubber Puncture Tester (D136-19-14).

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Level A. Four identical faceblanks shall be packaged in a style RSC fiberboard box conforming to W5C or W6C of PPP-B-636. The faceblanks shall be positioned in such a manner that a firm pack is obtained and that its natural form and contour will not be distorted. The box shall be closed and sealed in accordance with the appendix thereto.

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5.1.2 Level C. Faceblanks shall be packaged to provide adequate protection against deterioration and damage from the supply source to the first receiving activity for immediate use.

5.2 Packing.

5.2.1 Level A. Ten packages of faceblanks, packaged as specified in 5.1.1, shall be packed in a class 2, style 4, nailed wood box conforming to PPP-B-621 or in an overseas type, style A, cleated plywood box conforming to PPP-B-601 and using plywood conforming to group A of NN-P-530 in accordance with type I, grade 3 or 4, of CS-35 provided the plywood does not have continuous holes or openings extending the entire panel from face to back. The box shall be closed and strapped in accordance with the appendix thereto. Strapping conforming to type I, class B, of QQ-S-781 shall be used. The boxes and the finished wood parts thereof shall be completely immersed for a minimum of three minutes in a water-repellent wood preservative conforming to type II, composition A, of TT-W-572.

5.2.2 Level B. Ten packages of faceblanks, packaged as specified in 5.1.1, shall be packed in a style RSC fiberboard box conforming to V3c of PPP-B-636. The box shall be closed and sealed in accordance with the appendix thereto.

5.2.3 Level C. Faceblanks, packaged as specified in 5.1.2, shall be packed in a style RSC fiberboard box conforming to W5c or W6c of PPP-B-636. The box shall be closed and sealed in accordance with the appendix thereto.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages shall be marked in accordance with MIL-STD-129. Shipping containers shall be marked in accordance with drawing D150-2-1. All markings on the shipping containers shall be clear, legible, and permanent after the application of the wood preservative specified in 5.2.1.

5.4 Repair part. When this item is procured for storage and issue as a repair part, preservation, packaging, packing, and marking shall be as specified on the packaging data sheet which is identified by the Federal stock number.

6. NOTES

6.1 Intended use. The faceblank covered by this specification is intended for use in the manufacture of M9 Facepieces.

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6.2 Ordering data. Procurement documents should specify:

(a) Title, number, and date of this specification.
(b) Size of faceblank.
(c) Applicable drawings.
(d) Preproduction.

(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether or not to proceed with production.

6.3 Equivalent cure. Slabs and faceblanks of the same stock and gage are considered to have equivalent cures or the same state of vulcanization when:

(a) Their free-sulfur contents do not vary more than 10 percent.

(b) Their average tensile stress values at any given elongation (other than ultimate elongation) do not vary more than ± 2.5 percent.

(c) Their curing times can be equated in the following equation:

$$t_2 = t_1 \times 1.6^{\left(\frac{T_1 - T_2}{10}\right)}$$

where: t_1 = minutes of cure at temperature T_1,

t_2 = minutes of cure at temperature T_2,

T_1 = temperature (°F) of cure for time t_1, and

T_2 = temperature (°F) of cure for time t_2.

6.4 Major change. The term applies to any change in formulae, processing, or curing of currently approved rubber stocks which induce toxic characteristics and cause casualties. The major change should be considered effected when one or more of the following conditions occur:

(a) Components of formulae are varied in excess of the limitations stated in the manufacturer's tolerance (see 4.1.2.2).

(b) The addition or deletion of one or more components of formulae.

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(c) Substitutes for one or more components of formulae without advance approval of the Government.

(d) Deviation from the submitted approved factory mold cure exceeding /5°F or -10°F, even though equivalent cure is attained.

(e) Use of mold lubricants, dusting powders, or wet sprays not submitted with original information for toxicity approval.

6.5 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process and subjected to some physical mixing operation intended to make the final product substantially uniform.

6.6 Aging. Aging at 212° ± 1.8°F for a continuous period of 24 hours is slightly more severe on rubber than aging at 158° ± 1.8°F for 168 consecutive hours; therefore, specimens that fail to meet requirements when aged at the higher temperature may be discarded and replaced by new specimens aged at 158° ± 1.8°F for 168 consecutive hours.

6.7 Referee standards. During preproduction and subject to later revision during production, the Government will establish referee standards to define the type and category (major and minor) of each defect. This classification will be based upon the defect magnitude and potential deleterious effect on intended use.

Custodian:

Army - MU

Preparing activity:

Army - MU(EA)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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